

Optimization Methods

Basic concepts of optimization

Minima, gradients, Hessians, convexity, feasible directions

Today we build the vocabulary of the whole course

Everything we do later — gradient descent, Newton, constrained problems — is written in these five words.

SEMINAR GOALS

1 Define

local vs. global minimum, stationary point

2 Compute

gradients and Hessians without mistakes

3 Classify

stationary points: min, max, saddle

4 Recognise

convex sets and convex functions

5 Handle boundaries

feasible directions and the inequality form of optimality

TAKEAWAY

A stationary point is a *candidate*, not an answer. Most of today is about telling candidates apart.

The problem set for today

Eleven exercises in four blocks. Each one gets its own slide, and then its solution gets another.

BLOCK 1 · GRADIENTS AND HESSIANS

1 $f = x_1^2 + x_2^2 - x_1x_2$: find ∇f and H

2 $f = 3x_1^3 + 2x_1x_2^2 + 4x_2^2$ at $(2, -1)$

3 $f = x_1^2 + 2x_2^2 + 3x_3^2 + 2x_1x_2 - x_2x_3$

BLOCK 2 · CLASSIFYING STATIONARY POINTS

4 $f = x_1^2 + x_2^2$

5 $f = x_1^2 - x_2^2$

6 $f = x_1^3 - 3x_1x_2^2$ (monkey saddle)

7 $f = x_1^4 + x_2^4$

BLOCK 3 · CONVEXITY

8 Is the disk $x_1^2 + x_2^2 \leq 4$ convex?

9 Is $f = x_1^2 + x_2^2 - 4x_1x_2$ convex on $\mathbb{R}^2$?

BLOCK 4 · FEASIBLE DIRECTIONS

10 $\Omega = \{x_1 \geq 0, x_2 \geq 0\}$ at the point $(0, 2)$

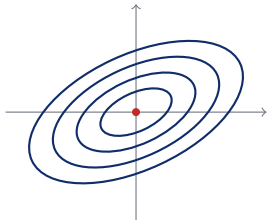
11 Directional derivative, two ways

HOW WE WORK

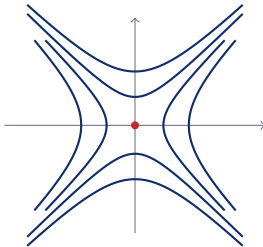
Three minutes on your own, in pairs if you prefer. Then we solve it together on the board. Interrupt me during the derivation — a question asked in the middle is worth far more than the same question afterwards.

Warm-up: read the picture before you differentiate

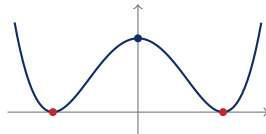
Look at the three landscapes and predict the answer. No calculus yet.



Closed elliptic level sets



Hyperbolic level sets



A one-dimensional slice

QUESTION

Which of the three marked points are local minima, which are global, and which piece of information — the gradient or the Hessian — did you actually use to decide?

Local and global minima: the definitions we will use all year

The only difference is the size of the neighbourhood.

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be defined on a region $\Omega \subset \mathbb{R}^n$.

LOCAL MINIMUM

$x^* \in \Omega$ is a local minimum if there exists $\varepsilon > 0$ such that

$$f(x) \geq f(x^*) \quad \text{for all } x \in \Omega \setminus \{x^*\}, \quad \|x - x^*\| < \varepsilon.$$

GLOBAL MINIMUM

$x^* \in \Omega$ is a global minimum if

$$f(x) \geq f(x^*) \quad \text{for all } x \in \Omega \setminus \{x^*\}.$$

Replacing $\geq$ by $>$ gives the *strict* versions.

TAKEAWAY

A global minimum is a local minimum whose neighbourhood is the whole of Ω . So every global minimum is local — and the converse fails, which is why the whole course exists.

Gradient and Hessian: notation and shapes

Fix the conventions now; a shape error is a mathematical error.

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right)^\top \in \mathbb{R}^n$$

$$H(x) = \left[\frac{\partial^2 f}{\partial x_i \partial x_j} \right]_{i,j=1}^n \in \mathbb{R}^{n \times n}$$

- H is symmetric whenever the second partials are continuous (Schwarz).
- For a quadratic function H is a constant matrix; in general it depends on x .

WORKED EXAMPLE FROM THE LECTURE

$$f(x) = 3x_1^3 + 2x_1x_2^2 + 4x_2^2 \text{ at } x = (1, 1).$$

$$\frac{\partial f}{\partial x_1} = 9x_1^2 + 2x_2^2, \quad \frac{\partial f}{\partial x_2} = 4x_1x_2 + 8x_2$$

$$H(x) = \begin{pmatrix} 18x_1 & 4x_2 \\ 4x_2 & 4x_1 + 8 \end{pmatrix}$$

$$H(1, 1) = \begin{pmatrix} 18 & 4 \\ 4 & 12 \end{pmatrix}, \quad \det H = 200$$

Block 1 · Exercise 1

Warm-up on a symmetric quadratic. Three minutes on your own.

PROBLEM

$$f(x_1, x_2) = x_1^2 + x_2^2 - x_1 x_2$$

1. Find ∇f at an arbitrary point.
2. Find the Hessian H .
3. Evaluate both at $(1, 1)$.
4. Before computing: will H depend on the point? Why?

HINT

Write the function as $\frac{1}{2}x^\top Ax$ and read off A . Which A do you get, and is it the matrix of coefficients you see in the formula?

Exercise 1 — solution

The Hessian of a quadratic is constant; the gradient is not.

$$\nabla f = \begin{pmatrix} 2x_1 - x_2 \\ 2x_2 - x_1 \end{pmatrix}, \quad H = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

$$\nabla f(1,1) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \neq 0, \quad H(1,1) = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

NOTE 1

$(1,1)$ is *not* stationary: $\nabla f \neq 0$ there, so it cannot be a minimum of an unconstrained problem.

NOTE 2

The cross term $-x_1x_2$ contributes -1 to *both* off-diagonal entries: $f = \frac{1}{2}x^\top Ax$ with $A = H$.

TAKEAWAY

Eigenvalues of H are 1 and 3: positive definite everywhere, so f is strictly convex and its unique minimum is at the origin.

Block 1 · Exercise 2

Same routine, but now the Hessian moves with the point.

PROBLEM

$$f(x_1, x_2) = 3x_1^3 + 2x_1x_2^2 + 4x_2^2 \quad \text{at } x = (2, -1)$$

1. Compute $\nabla f(2, -1)$.
2. Compute $H(2, -1)$ and its determinant.
3. Compare with the lecture, where the same function was evaluated at $(1, 1)$.
4. Is $(2, -1)$ a stationary point?

WATCH OUT

Signs. Two of the four entries change sign when x_2 becomes negative.

Exercise 2 — solution

A cubic term makes curvature a local property.

$$\frac{\partial f}{\partial x_1} = 9x_1^2 + 2x_2^2 = 9 \cdot 4 + 2 \cdot 1 = 38, \quad \frac{\partial f}{\partial x_2} = 4x_1x_2 + 8x_2 = -8 - 8 = -16$$

$$H(x) = \begin{pmatrix} 18x_1 & 4x_2 \\ 4x_2 & 4x_1 + 8 \end{pmatrix} \Rightarrow H(2, -1) = \begin{pmatrix} 36 & -4 \\ -4 & 16 \end{pmatrix}, \quad \det H = 576 - 16 = 560$$

ANSWER TO (4)

$\nabla f(2, -1) = (38, -16)^\top \neq 0$, so the point is not stationary and the Hessian says nothing about optimality here.

ANSWER TO (3)

At $(1, 1)$ we got $\det H = 200$, here 560. For non-quadratic f , curvature is a *local* object — it changes from point to point.

Block 1 · Exercise 3

Three variables, and our first use of Sylvester's criterion.

PROBLEM

$$f(x_1, x_2, x_3) = x_1^2 + 2x_2^2 + 3x_3^2 + 2x_1x_2 - x_2x_3$$

1. Find ∇f and H .
2. Decide whether H is positive definite.
3. What does your answer imply about convexity of f on $\mathbb{R}^3$?
4. Where is the minimum, and is it unique?

HINT

A cross term $c x_i x_j$ with $i \neq j$ puts c into *both* entries H_{ij} and H_{ji} — the Hessian entry is the second derivative itself. (The matrix A in $f = x^T A x$ would get $c/2$; do not mix the two conventions.)

Exercise 3 — solution

Leading principal minors decide definiteness without eigenvalues.

$$\nabla f = \begin{pmatrix} 2x_1 + 2x_2 \\ 4x_2 + 2x_1 - x_3 \\ 6x_3 - x_2 \end{pmatrix}, \quad H = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 4 & -1 \\ 0 & -1 & 6 \end{pmatrix} \quad (\text{constant})$$

$$\Delta_1 = 2 > 0, \quad \Delta_2 = 2 \cdot 4 - 2 \cdot 2 = 4 > 0, \quad \Delta_3 = \det H = 22 > 0$$

Expansion along the first row: $\Delta_3 = 2(4 \cdot 6 - 1) - 2(2 \cdot 6 - 0) + 0 = 46 - 24 = 22$.

So $H \succ 0$: the function is strictly convex on $\mathbb{R}^3$, the stationary point $\nabla f = 0$ is unique and is the global minimum; solving the linear system gives $x^* = 0$.

TAKEAWAY

For a quadratic function, “strictly convex” and “unique global minimum” are the same statement about one constant matrix.

Sylvester's criterion — and the trap inside it

It answers “definite or not”. It does not answer “semidefinite or not”.

Let Δ_k be the k -th *leading* principal minor of a symmetric A .

THE CRITERION

$A \succ 0 \iff \Delta_k > 0$ for all k .

$A \prec 0 \iff (-1)^k \Delta_k > 0$ for all k , i.e. signs alternate starting with $-$.

Otherwise: indefinite *or* semidefinite — the leading minors cannot tell which.

COUNTEREXAMPLE TO REMEMBER

$$A = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} : \quad \Delta_1 = 0, \quad \Delta_2 = 0$$

Both leading minors are non-negative, yet A is negative semidefinite. For semidefiniteness you must check *all* principal minors, not only the leading ones.

PRACTICAL RULE

On paper use Sylvester for strict definiteness. In code use `numpy.linalg.eigvalsh` with a tolerance — it is the honest tool for the borderline cases.

Optimality conditions: the summary table

Necessary conditions filter candidates. Only the sufficient condition certifies an answer.

	Necessary	Sufficient
First order	$\nabla f(x^*) = 0$ (interior point)	—
Second order	$H(x^*) \succeq 0$	$\nabla f(x^*) = 0$ and $H(x^*) \succ 0$

AT THE BOUNDARY

The equality weakens to an inequality:
 $s^\top \nabla f(x^*) \geq 0$ for every feasible direction s .
That inequality is the seed of KKT theory
later in the course.

THE GAP

Between $\succeq 0$ and $\succ 0$ lives everything
interesting: flat directions, where the
second-order test simply has no opinion.
Exercises 6 and 7 live exactly there.

Block 2 · Exercise 4

The simplest landscape there is. Two minutes.

PROBLEM

$$f(x_1, x_2) = x_1^2 + x_2^2$$

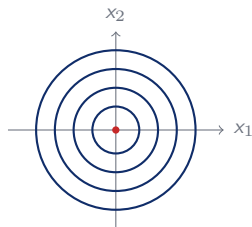
1. Find all stationary points.
2. Compute the Hessian and decide its definiteness.
3. Which of the two conditions — necessary or sufficient — did you just use?
4. Is the point a *global* minimum, and what exactly justifies that?

WATCH THE WORDING

“The Hessian is positive definite” certifies a *local* minimum. Global needs one extra sentence — find it.

Exercise 4 — solution

A bowl: positive curvature in every direction.



Level sets are circles

$$\nabla f = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix} = 0 \Rightarrow x^* = (0, 0)$$

$$H = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad \Delta_1 = 2 > 0, \quad \Delta_2 = 4 > 0$$

$H \succ 0$, so the sufficient condition applies: $(0, 0)$ is a *strict local* minimum. Global follows separately — either from convexity ($H \succeq 0$ everywhere) or directly from $f(x) = x_1^2 + x_2^2 \geq 0 = f(0, 0)$.

TAKEAWAY

Eigenvalues 2 and 2: the same curvature in every direction, which is why the level sets are perfect circles and why gradient descent would land here in a single step.

Block 2 · Exercise 5

Same gradient structure, opposite conclusion. Two minutes.

PROBLEM

$$f(x_1, x_2) = x_1^2 - x_2^2$$

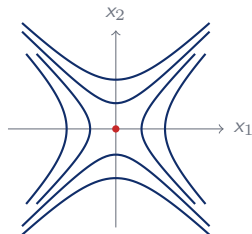
1. Show that $(0, 0)$ is a stationary point.
2. Show that it is neither a minimum nor a maximum.
3. Do it *without* drawing a picture — a picture is a guide, not a proof.

HINT

Restrict the function to a line through the origin. Two well-chosen lines settle the question completely.

Exercise 5 — solution

Mixed curvature signs always mean a saddle.



Level sets are hyperbolas

$$\nabla f = \begin{pmatrix} 2x_1 \\ -2x_2 \end{pmatrix} = 0 \Rightarrow (0, 0)$$

$$H = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}, \quad \Delta_1 = 2 > 0, \quad \Delta_2 = -4 < 0$$

Neither Sylvester pattern holds: the matrix is indefinite, with eigenvalues 2 and -2 . The proof without pictures: $f(x_1, 0) = x_1^2 > 0$ and $f(0, x_2) = -x_2^2 < 0$, so every neighbourhood of the origin contains larger *and* smaller values.

TAKEAWAY

The gradient locates candidates; the eigenvalue *signs* classify them. Mixed signs always mean a saddle.

Block 2 · Exercise 6: the monkey saddle

Where the second-order test runs out of information.

PROBLEM

$$f(x_1, x_2) = x_1^3 - 3x_1x_2^2$$

1. Find all stationary points.
2. Write the Hessian and evaluate it at the origin.
3. Try to classify the point with the tools from the table. What happens?
4. Find a direction that settles the question.

HINT

When the Hessian is the zero matrix, stop using theorems and start using the *definition*: compare f with $f(x^*)$ along a cleverly chosen line.

Exercise 6 — solution

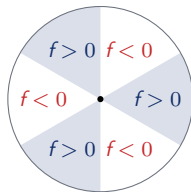
The name comes from the three valleys: two for the legs, one for the tail.

$$\frac{\partial f}{\partial x_1} = 3x_1^2 - 3x_2^2, \quad \frac{\partial f}{\partial x_2} = -6x_1x_2$$

From the second equation $x_1 = 0$ or $x_2 = 0$; the first one then leaves only $(0, 0)$.

$$H(x) = \begin{pmatrix} 6x_1 & -6x_2 \\ -6x_2 & -6x_1 \end{pmatrix} \Rightarrow H(0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

The zero matrix is both $\succeq 0$ and $\preceq 0$: the necessary condition holds trivially, the sufficient one does not apply, and all $\Delta_k = 0$, so Sylvester is silent. Along $x_2 = 0$ we get $f(x_1, 0) = x_1^3$, which changes sign at the origin: the point is neither a minimum nor a maximum.



Six alternating sectors around the origin

Block 2 · Exercise 7

One more zero Hessian — and this time the answer is different. Two minutes.

PROBLEM

$$f(x_1, x_2) = x_1^4 + x_2^4$$

1. Find the stationary points and the Hessian at the origin.
2. Try the second-order test. What does it say?
3. Decide the question anyway — in one line.
4. Compare your answer with Exercise 6. What is different, and what is identical?

HINT

You do not need any theorem here. Look at the sign of the function itself.

Exercise 7 — solution

Identical first- and second-order data, opposite answers.

$$\nabla f = (4x_1^3, 4x_2^3)^\top = 0 \Rightarrow (0, 0), \quad H(x) = \begin{pmatrix} 12x_1^2 & 0 \\ 0 & 12x_2^2 \end{pmatrix} \Rightarrow H(0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Direct argument: $f(x) = x_1^4 + x_2^4 \geq 0 = f(0, 0)$, with equality only at the origin, so the origin is a *strict global* minimum — no theorem required.

Function	$\nabla f(0)$	$H(0)$	Truth
$x_1^4 + x_2^4$	0	0	strict global minimum
$x_1^3 - 3x_1x_2^2$	0	0	neither minimum nor maximum

TAKEAWAY

Same data, opposite answers. When the Hessian is singular, the second-order test is not “inconclusive by bad luck” — it provably cannot decide.

Block 3 · Convexity: definitions

The property that turns a local statement into a global one.

CONVEX SET

Ω is convex if for any $x_1, x_2 \in \Omega$ the whole segment $[x_1, x_2]$ lies in Ω :

$$\lambda x_1 + (1 - \lambda)x_2 \in \Omega, \quad \lambda \in [0, 1]$$

Geometrically: the graph never rises above the chord connecting two of its points. For $f \in \mathcal{C}^2$ this is equivalent to $H(x) \succeq 0$ for every $x \in \Omega$.

CONVEX FUNCTION

f is convex on a convex Ω if for all $x, y \in \Omega$, $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

WHY WE CARE

For a convex f on a convex Ω , every local minimum is global, and $\nabla f(x^*) = 0$ stops being merely necessary — it becomes sufficient.

Block 3 · Exercise 8

A proof straight from the definition. Two minutes.

PROBLEM

$$\Omega = \{(x_1, x_2) : x_1^2 + x_2^2 \leq 4\}$$

1. Prove from the definition that Ω is convex.
2. Which properties of the norm did your proof use?
3. Does the same argument work for a square? For a ring?

HINT

Rewrite the set as $\{x : \|x\| \leq 2\}$. Then take $z = \lambda x + (1 - \lambda)y$ and estimate $\|z\|$ — two standard inequalities are enough.

Exercise 8 — solution

Convexity of any ball is the triangle inequality in disguise.

Take $x, y \in \Omega$, so $\|x\| \leq 2$ and $\|y\| \leq 2$, and take $\lambda \in [0, 1]$. Put $z = \lambda x + (1 - \lambda)y$. Then

$$\|z\| = \|\lambda x + (1 - \lambda)y\| \stackrel{(1)}{\leq} \|\lambda x\| + \|(1 - \lambda)y\| \stackrel{(2)}{=} \lambda\|x\| + (1 - \lambda)\|y\| \leq 2\lambda + 2(1 - \lambda) = 2,$$

where (1) is the triangle inequality and (2) is homogeneity of the norm (the factors are non-negative, so no absolute values are needed). Hence $z \in \Omega$, and Ω is convex. ■

HOW FAR IT GENERALISES

Nothing but two norm properties was used, so every ball in every norm is convex — including the ℓ_1 diamond and the ℓ_∞ square, which have corners. A ring is not convex: the segment between two opposite points crosses the hole.

Block 3 · Exercise 9

Convexity of a quadratic, decided by one matrix. Two minutes.

PROBLEM

$$f(x_1, x_2) = x_1^2 + x_2^2 - 4x_1x_2$$

1. Is f convex on $\mathbb{R}^2$?
2. Compare with Exercise 1, where the cross term was $-x_1x_2$. What changed?
3. For which a is $x_1^2 + x_2^2 + a x_1x_2$ convex?

HINT

For a quadratic the Hessian is constant, so convexity is a single yes/no question about a single matrix.

Exercise 9 — solution

It is not the sign of the cross term that matters — it is its size.

$$H = \begin{pmatrix} 2 & -4 \\ -4 & 2 \end{pmatrix} \text{ (constant)}, \quad \Delta_1 = 2 > 0, \quad \Delta_2 = 4 - 16 = -12 < 0$$

Neither Sylvester pattern holds, so H is indefinite — eigenvalues 6 and -2 . Since H is the same at every point, it is nowhere $\succeq 0$, and therefore f is *not* convex on $\mathbb{R}^2$.

For the whole family $f_a = x_1^2 + x_2^2 + a x_1 x_2$ the Hessian is $\begin{pmatrix} 2 & a \\ a & 2 \end{pmatrix}$ with eigenvalues $2 \pm a$:

$$\text{convex} \iff |a| \leq 2, \quad \text{strictly convex} \iff |a| < 2.$$

TAKEAWAY

Exercise 1 had $a = -1$ (eigenvalues 1, 3, convex); here $a = -4$ (eigenvalues $-2, 6$, not convex). The cross term competes with the squares, and at $|a| = 2$ it exactly ties: $f = (x_1 \pm x_2)^2$ has a whole line of minima.

Block 4 · Feasible directions and directional derivatives

What optimality means when you are standing on the boundary.

FEASIBLE DIRECTION

$s \neq 0$ is feasible at $x \in \Omega$ if there is $\alpha_0 > 0$ with $x + \alpha s \in \Omega$ for all $\alpha \in [0, \alpha_0]$.

DIRECTIONAL DERIVATIVE

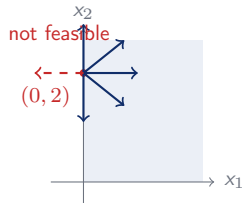
$$\frac{\partial f}{\partial s}(x) = \lim_{\alpha \rightarrow 0} \frac{f(x + \alpha s) - f(x)}{\alpha} = s^\top \nabla f(x)$$

If $\|s\| = 1$ this is the rate of increase of f along s .

FIRST-ORDER NECESSARY CONDITION

$$s^\top \nabla f(x^*) \geq 0 \quad \text{for every feasible } s$$

Interior point: every direction is feasible, so $\pm s$ both qualify and this collapses to $\nabla f(x^*) = 0$.



Feasible cone at $(0, 2)$ in the first quadrant

Block 4 · Exercise 10

Optimality on the boundary. Three minutes.

PROBLEM

$$\Omega = \{x_1 \geq 0, x_2 \geq 0\}, \quad x = (0, 2), \quad f(x_1, x_2) = x_1 + x_2$$

1. Which constraints are active at $(0, 2)$?
2. Describe the full set of feasible directions there.
3. Check the first-order necessary condition $s^\top \nabla f \geq 0$.
4. If it fails, produce the direction that breaks it.

HINT

One coordinate is pinned at its bound, the other has room on both sides. That asymmetry is the whole exercise.

Exercise 10 — solution

One bad direction is enough to disqualify a point.

$\Omega = \{x_1 \geq 0, x_2 \geq 0\}$, the point $x = (0, 2)$, the objective $f(x_1, x_2) = x_1 + x_2$.

- $x_1 = 0$ is active, $x_2 = 2 > 0$ is not. Hence the feasible set of directions is

$$\{s = (s_1, s_2) \neq 0 : s_1 \geq 0, s_2 \in \mathbb{R}\}.$$

- $\nabla f \equiv (1, 1)^\top$, so the condition to test is $s^\top \nabla f = s_1 + s_2 \geq 0$ for every feasible s .
- Take $s = (0, -1)$: feasible, and $s^\top \nabla f = -1 < 0$.

Therefore $(0, 2)$ fails the first-order necessary condition and is not a candidate for a local minimum — consistent with the true minimiser $(0, 0)$.

TAKEAWAY

To *reject* a point you need one violating direction. To *accept* it you must check all of them — which is precisely why we will later replace this test by KKT multipliers.

Block 4 · Exercise 11

Definition versus formula. Three minutes.

PROBLEM

$$f(x_1, x_2) = x_1^2 + 2x_2^2, \quad x = (1, 1), \quad s = (1, -1)$$

1. Compute $\frac{\partial f}{\partial s}(x)$ from the definition, through the limit.
2. Compute it again as $s^\top \nabla f(x)$.
3. Is s a descent direction?
4. Careful: is your number a rate per unit of α , or per unit of length?

HINT

For part 1 substitute $x + \alpha s$ into f , expand, and let $\alpha \rightarrow 0$. You should get a quadratic in α whose constant term is $f(x)$.

Exercise 11 — solution

The formula and the definition must agree; verifying that is the point.

$$f(x_1, x_2) = x_1^2 + 2x_2^2 \text{ at } x = (1, 1) \text{ along } s = (1, -1).$$

WAY 1: FROM THE DEFINITION

$$\begin{aligned} f(x + \alpha s) &= (1 + \alpha)^2 + 2(1 - \alpha)^2 \\ &= 3 - 2\alpha + 3\alpha^2 \end{aligned}$$

$$\frac{\partial f}{\partial s} = \lim_{\alpha \rightarrow 0} \frac{(3 - 2\alpha + 3\alpha^2) - 3}{\alpha} = -2$$

WAY 2: FROM THE GRADIENT

$$\nabla f = (2x_1, 4x_2)^\top \Rightarrow \nabla f(1, 1) = (2, 4)^\top$$

$$s^\top \nabla f = 1 \cdot 2 + (-1) \cdot 4 = -2$$

Both routes give -2 , so $(1, -1)$ is a descent direction at $(1, 1)$.

NOTE ON NORMALISATION

$\|s\| = \sqrt{2} \neq 1$, so -2 is a rate per unit of α , not per unit of length: per unit length it is $-2/\sqrt{2} = -\sqrt{2}$.

From the board to the keyboard

Every statement of today has a three-line numerical counterpart. Use them in the homework.

```
import numpy as np

def central_diff(f, x, h=1e-5):
    x = np.asarray(x, float)
    g = np.empty_like(x)
    for j in range(x.size):
        e = np.zeros_like(x); e[j] = h
        g[j] = (f(x + e) - f(x - e)) / (2*h)
    return g
```

```
def classify(H, tol=1e-8):
    w = np.linalg.eigvalsh(0.5*(H + H.T))
    if np.all(w > tol): return "min"
    if np.all(w < -tol): return "max"
    if w.min() < -tol < w.max():
        return "saddle"
    return "inconclusive"
```

QUESTION

Run `classify` on the Hessians of $x_1^4 + x_2^4$ and $x_1^3 - 3x_1x_2^2$ at the origin. Both return "inconclusive". Which of the two answers is a limitation of the code, and which is a limitation of the mathematics?

Summary

Five sentences to carry into the next lecture.

1. A global minimum is a local minimum with the largest possible neighbourhood; the whole difficulty of optimization lives in the gap between the two.
2. The gradient supplies candidates, the Hessian classifies them — and only for interior points of Ω .
3. Necessary conditions cannot certify a minimum; only $\nabla f = 0$ together with $H \succ 0$ can.
4. When H is singular the second-order test is genuinely undecided: $x_1^4 + x_2^4$ and $x_1^3 - 3x_1x_2^2$ are indistinguishable to it, yet one is a minimum and the other is not.
5. On the boundary the equality $\nabla f = 0$ becomes the inequality $s^\top \nabla f \geq 0$ over feasible directions — the starting point of constrained optimization.

Exit ticket and home practice

Two minutes of writing now; four exercises before the next seminar.

EXIT TICKET (ANSWER IN ONE LINE EACH)

- 1 Give a function whose Hessian is positive semidefinite at a point that is not a local minimum.
- 2 Why does the first-order condition become an inequality at the boundary?
- 3 A colleague reports $\Delta_1 > 0$ and $\Delta_2 = 0$. What can you conclude?

HOME PRACTICE

- 1 Classify the stationary points of $f = x_1^3 + x_2^2$.
- 2 For which a is $x_1^2 + x_2^2 + a x_1 x_2$ convex on $\mathbb{R}^2$?
- 3 Describe the feasible directions at $(0, 0)$ for $\Omega = \{x_1 \geq 0, x_2 \geq 0\}$.
- 4 Implement `classify` and test it on all functions from today; report where it says "inconclusive".

NEXT TIME

We start descending: how to move from a point that is not optimal to one that is, and how fast that is possible.